NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

Rapid Trials Lab mainline

Group Rapid Trials Lab

Status Promoted

Period 2026-10-15 23:46 — 2026-11-13 19:25

Stage conclusion

First choice is K-Dense, with Academic as a style reference; they handle only styling and export: layout must be written by hand and structure checked by a person

Project report

Redrawing the Graph-DeepONet final network architecture diagram with scientific plotting Skills

Objective

Verify the repositories and licenses of 6 candidate Skills (AgentFigureGallery, Nature Skill, Academic Figure Skill, Origin Style Replot, figures4papers, K-Dense Scientific Visualization), select at most 2 (prioritizing those that output editable files) to test-draw the "final network architecture diagram" of the openclaw urban wind-field Graph-DeepONet project, and render a worth-using / drop verdict.

How the test target was selected

Candidates: the 04-09 figure (UNet Speed Distillation Round 34, Global Best; boxes and connector lines overlap), checked against later final versions in the same folder: the 7-level UNet final diagram (04-15) and the project notes (recording the best model as HybridGraphCNN+FineGraph48+Fourier64, with no existing diagram). Which was chosen and why: both the round34 original figure (04-09) and the 7-level UNet final diagram (04-15) belong to a separate UNet reproduction track (`auto_v2`, Lu et al. 2025) and do not depict Graph-DeepONet. The optimal architecture of Graph-DeepONet exists only in text notes (04-02), where lines 35–36 specify a HybridGraphCNN + FineGraph48 + Fourier64 trunk; therefore a schematic was drawn from the text record, with any unrecorded layer counts or channel dimensions explicitly labeled "unrecorded" (original text verified by the lead on 10-16). To answer the question the PI raised ("can the overlaps in the original figure be fixed?"), a supplementary test also redrew the round34 original figure.

Current status

10-16 03:20 Wrap-up: Worth using (with limitations), per the headline conclusion. Graph-DeepONet has no existing final architecture diagram, only text records; the schematic drawn from those records is labeled "not verified against code" and cannot be used as a final journal submission figure.

Security check

See the audit report. Correction by the lead: the `eval(parse(...))` call in Academic's compose script appears only inside a docstring usage example and is never executed anywhere in the repository (verified via `grep`), representing low risk and not grounds for disqualification.

Trials and results

Date	Action taken	One-line result
10-16	Supplementary test: redrew the round34 original figure with both Skills	Verified by the lead: title, subtitle, numbers across all 9 modules, 3 skip connections, and auxiliary head all match the original figure; all box-on-box overlaps and curve-through-box crossings in the original are eliminated; each SVG has 38 editable text elements and the PDFs have embedded fonts
10-16	Audited 6 repositories: 5 located, no matching repository found for Origin Style Replot; figures4papers is CC BY-NC	Selected Academic Figure Skill (Apache-2.0) and K-Dense `scientific-visualization` (MIT)

Date	Action taken	One-line result
10-16	Drew Graph-DeepONet schematic from text records	Both figures are consistent in content, overlap-free, and have editable SVG text; the K-Dense version has excessive outer margins; all unrecorded parameters are explicitly labeled without fabricated numbers

Open items and next steps

- Hand implementation recommendation to Platform Lead: equip the plotting bot (`figures`) only with the `scientific-visualization` subdirectory of K-Dense pinned to commit `a1043ef544`, without installing the full 177-skill repository; from Academic, take only the style reference documents under `references`. Delivery gate: item-by-item checklist against the original figure/notes + visual PNG inspection + editable SVG text.
- If a final submission figure for Graph-DeepONet is needed: first obtain the source code or configuration of the best model to verify the exact layer counts and channel dimensions.

Key figures

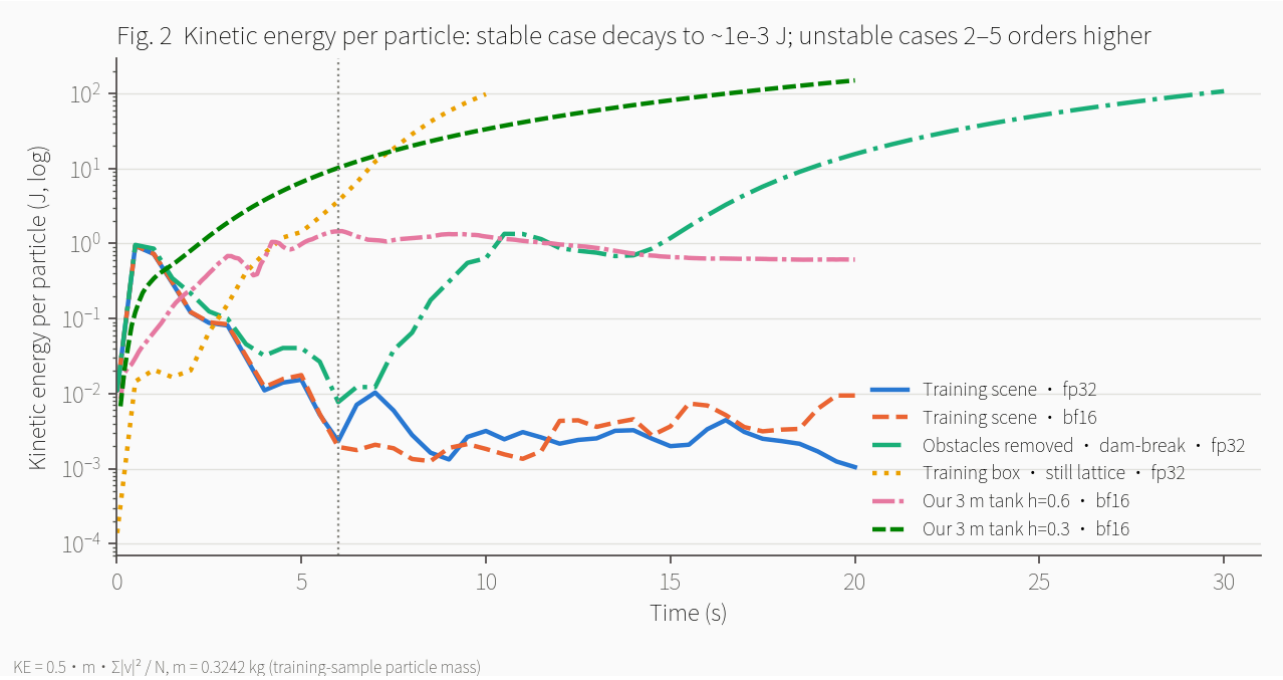


fig2 kinetic energy

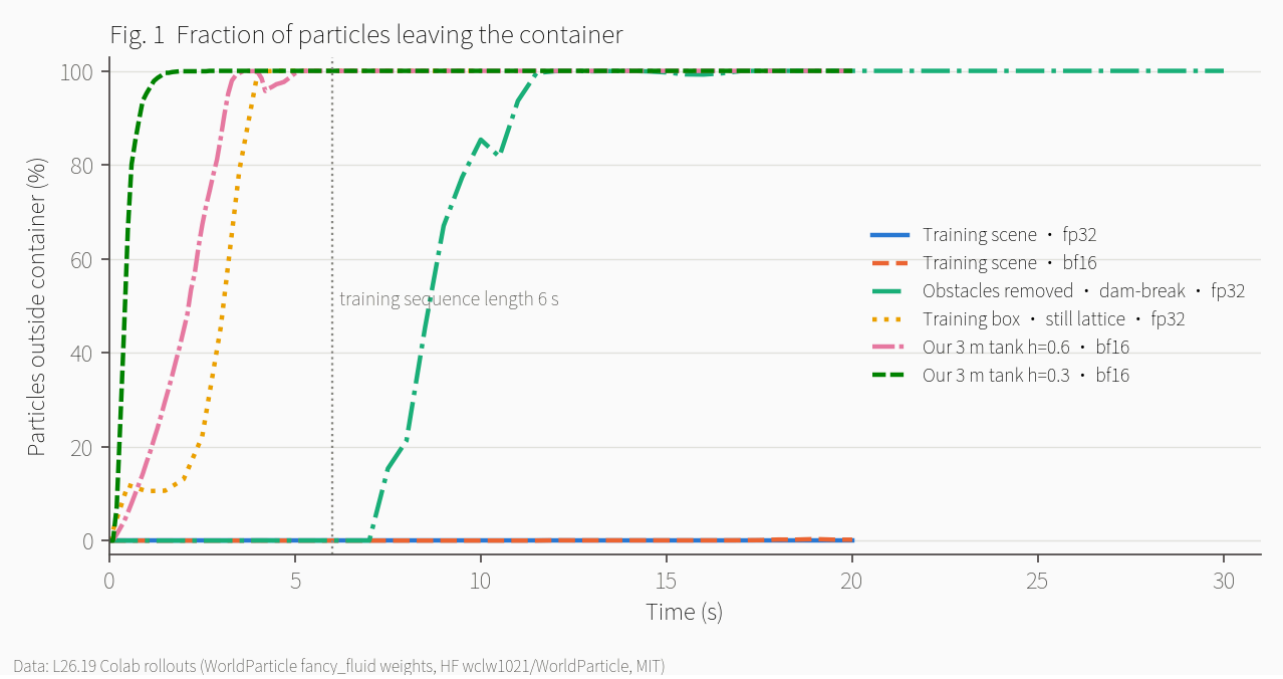
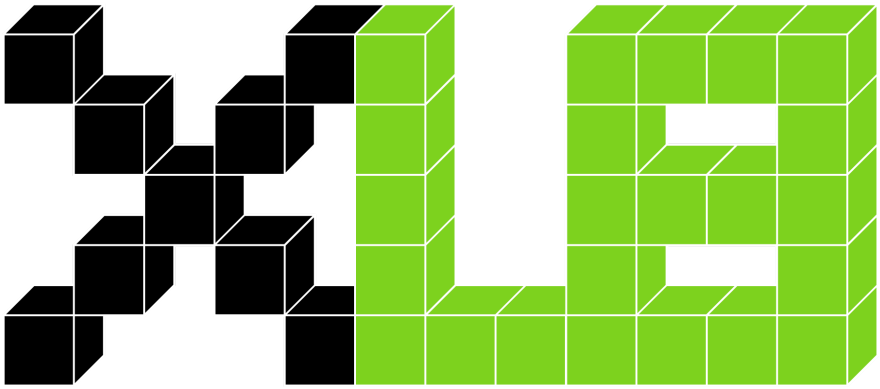
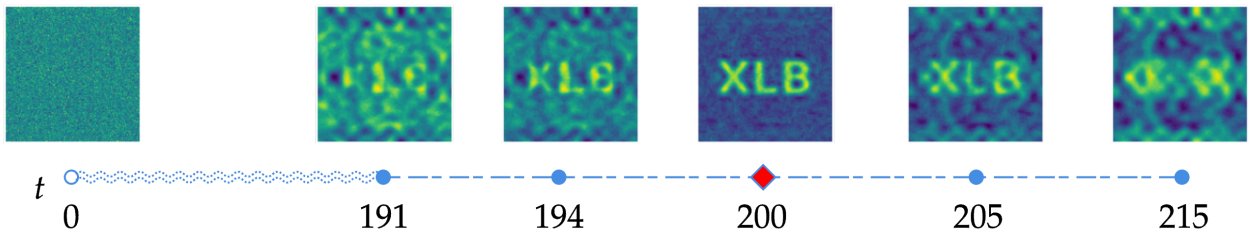


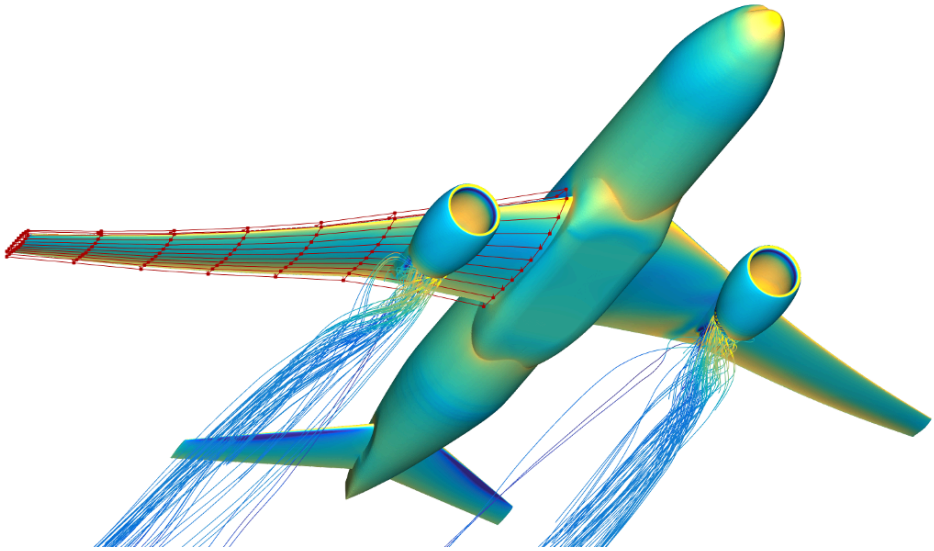
fig1 out of bounds



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XLB diff



cover